



In situ sulfide saturation induced by geometric changes of magma conduit in the world-class Jinchuan magmatic Ni-Cu sulfide deposit, China

Qing-Han Yuan^{a,b,*}, Ben-Xun Su^{a,b}, Meng-Meng Cui^a

^a Key Laboratory of Mineral Resources, Institute of Geology and Geophysics, Chinese Academy of Sciences, Beijing 100029, China

^b University of Chinese Academy of Sciences, Beijing 100049, China

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ABSTRACT

When and how sulfide saturation occurred in the Jinchuan magmatic Ni-Cu sulfide deposit are poorly constrained. To address this issue, we investigated compositional variations and crystallization temperatures of olivine in various ores and rocks in orebody 1 of the Jinchuan deposit. The forsterite (Fo) values and Mn contents in olivine vary respectively from 79.6 to 85.2 and 1270 to 2463 ppm in net-textured ores, 76.7 to 84.2 and 1425 to 2865 ppm in patchy net textured ores, 78.7 to 84.1 and 1440 to 2649 ppm in disseminated ores, and 69.7 to 81.8 and 1495 to 3191 ppm in sulfide-barren lherzolites. The estimated crystallization temperatures of olivine in these ores/rocks range from 803 to 1163 °C, 871 to 1314 °C, 821 to 1507 °C, and 929 to 1652 °C, respectively. These data revealed two distinct groupings in both olivine Fo-Mn correlations and crystallization temperatures cluster splitting at a Fo of 79.5 and a temperature of 1100 °C. The wider range of Mn content with a given Fo value higher than 79.5 compared to those with Fo values lower than 79.5 indicates the onset of in situ sulfide saturation at a Fo of 79.5. The termination of sulfide saturation is epitomized by olivine in disseminated ores with a peak Fo content of 84. The abrupt temperature drops from ca. 1400–1500 °C to ca. 800–1100 °C in the parental magma likely occurred when the magma flowed through the westernmost portion of this segment, which resembles a de Laval nozzle. This geometry could have induced rapid cooling and decompression of the magma, consequently promoting degassing through decompression. These reductions in temperature and H₂O content of the magma would collectively result in a decrease in the sulfur concentration at sulfide saturation (SCSS), which would exceed any SCSS increase due to decompression. Our findings emphasize the critical role of magma conduit geometry in physicochemical changes of the parental magma for sulfide saturation and subsequent mineralization in the Jinchuan deposit, which may also be applicable to other magmatic deposit worldwide.

1. Introduction

Sulfide saturation is a critical step in the formation of magmatic Ni-Cu sulfide deposits, as it governs sulfide segregation and concentrations of chalcophile elements from the silicate magma (e.g., Naldrett, 1997; Ripley and Li, 2013). This process is dependent on sulfur contents of magmas and threshold of sulfur concentration at sulfide saturation (SCSS), that is, sulfide saturation could occur when sulfur content of magmas exceeds the SCSS (e.g., Fortin et al., 2015; Wykes et al., 2015). The introduction of external sulfur is an effective means to surpass the SCSS (e.g., Ripley and Li, 2013), a process reflected in the mixed mantle source and country rock sulfur isotope ($\delta^{34}\text{S}$) signatures of deposits like the world-class Noril'sk in Russia (e.g., Li et al., 2009) and Voisey's Bay in Canada (e.g., Ripley et al., 1999). In contrast, the Jinchuan deposit in

China, the largest known single magmatic Ni-Cu deposit (Li and Ripley, 2011), exhibits $\delta^{34}\text{S}$ values (−7.6 to 3 ‰) that deviate from a mix of mantle sulfur (0 ± 2 ‰) and country rock sulfur (−4 to 11.3 ‰) (Xue et al., 2023), suggesting limited external sulfur influence and implying that sulfide saturation here may stem from a reduction in the SCSS itself. The determination of the SCSS is not straightforward, as it could be affected by physicochemical conditions, including magma composition (e.g., Li and Ripley, 2005, 2009; Wykes et al., 2015), oxygen fugacity (Jugo et al., 2005; Jugo, 2009), temperature and pressure conditions (e.g., Mavrogenes and O'Neill, 1999; Ding et al., 2018). Therefore, pinpointing the relative timing when sulfide saturation occurs is crucial for understanding the multitude of factors that lead to sulfide saturation within a magmatic system.

Sulfide saturation and subsequent removal of sulfide liquids from the

* Corresponding author.

E-mail address: yuanqinghan@mail.iggcas.ac.cn (Q.-H. Yuan).

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magma can lead to the depletion of Fe in the immiscible silicate melt, as Fe is preferentially partitioned into the sulfide (e.g., [Brenan, 2003](#)). Fe depletion can consequently result in a relative enrichment of Mg and Mn in the silicate melt. Olivine grains crystallizing from this Fe-depleted and Mg- and Mn-enriched melt will record these compositional changes. Therefore, the in situ crystallized olivine of the orebody 1 ([Mao et al., 2018](#)) may present an opportunity to track history of the sulfide saturation process.

In this study, we examined chemical variations in olivine, particularly Fo and Mn contents to elucidate the relative timing of sulfide saturation within the orebody 1 of the Jinchuan deposit. By pinpointing the onset of sulfide saturation and its interconnection with the crystallization of silicate minerals, we aim to shed light on triggers of sulfide saturation.

2. Geology of the Jinchuan deposit and sampling

The Jinchuan deposit, the third-largest magmatic Ni-Cu sulfide deposit in the world ([Naldrett, 2011](#)), is situated in the Longshoushan terrane ([Fig. 1a](#)), which is located on the western margin of the North China craton. The Longshoushan terrane is primarily composed of Proterozoic metasedimentary and metavolcanic rocks, as well as Archean basement rocks ([Fig. 1a](#); [Sixth Geological Unit, 1984](#)). The Jinchuan intrusion, which hosts the Ni-Cu sulfide mineralization, was emplaced

into the Proterozoic Baijiazui Formation, a sequence of metasedimentary rocks including marbles, gneisses and schists ([Fig. 1b](#); [Sixth Geological Unit, 1984](#); [Chai and Naldrett, 1992a](#)) at ca. 832 Ma ([Zhang et al., 2010](#)). The Jinchuan intrusion extends approximately 6.5 km in length, varies from 20 to about 500 m in width, and persists down-dip for more than 1100 m ([Fig. 1c](#)). It is primarily composed of dunite, lherzolite, plagioclase lherzolite and olivine pyroxenite with olivine, orthopyroxene, clinopyroxene, spinel, phlogopite, and amphibole as rock-forming minerals ([Sixth Geological Unit, 1984](#)). The ores exhibit a range of textures, from massive to net-textured to disseminated, with decreasing volume percentages of interstitial sulfide: > 90 vol% in massive ores, 15–35 vol% in net-textured ores, and 7–15 vol% in disseminated ores. Patchy net-textured ores, a sub-type of net-textured ores, have a sulfide volume percentage between that of net-textured and disseminated ores ([Sixth Geological Unit, 1984](#)).

The Jinchuan deposit was dislocated by SW-NE trending strike-slip faults and divided into a series of four mining segments designated as III, I, II and IV, arranged from west to east ([Fig. 1b-c](#)). The segments III and I each comprises two separate units: a sulfide-free Upper unit consisting of fine-grained dunite and lherzolite with minor pyroxenite, and sulfide-bearing Lower unit composed of coarse-grained lherzolite and dunite (e.g., [Song et al., 2012](#)). East of the F_{16-1} fault, the segments II and IV are characterized by a concentric distribution of rock types with a core of sulfide-bearing dunite that is enveloped by lherzolite (e.g., [Chai](#)

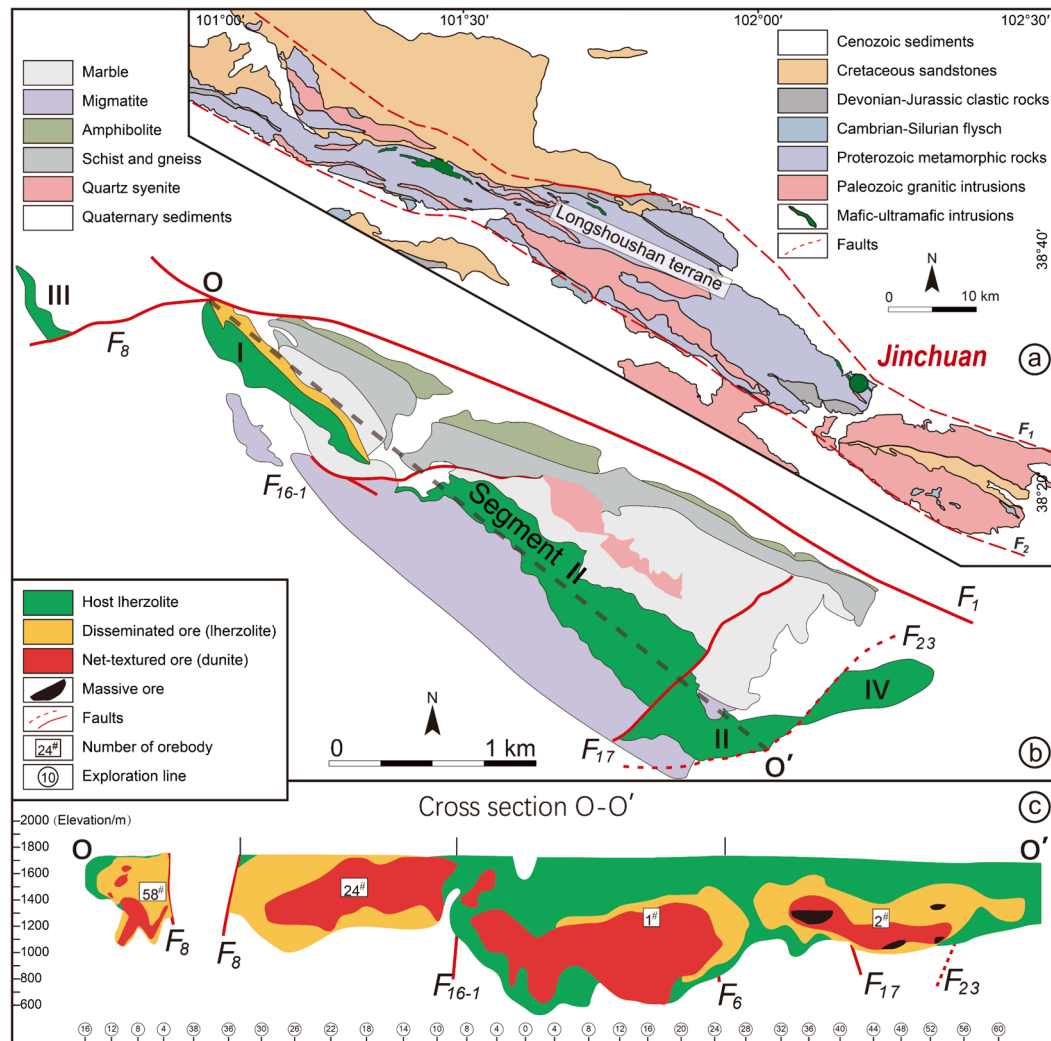


Fig. 1. Geological maps of (a) the Longshoushan terrane with the location of Jinchuan intrusion, (b) the Jinchuan intrusion and its country rocks, and (c) longitudinal cross section of Segments III, I and II (modified after [Li and Ripley, 2011](#)).

and Naldrett, 1992a; Song et al., 2012). The west sub-chamber of the segment II features a core of net-textured sulfide-bearing dunite surrounded by zone disseminated sulfide-bearing dunite and lherzolite. These sulfide-bearing rocks comprise orebody 1 (Fig. 1c), which is the most valuable orebody of the Jinchuan deposit. The samples in this study, including (patchy) net-textured and disseminated ores and sulfide-barren lherzolite, were collected from underground adits at the 1018 m, 934 m, and 814 m levels in orebody 1, and their mineralogy and texture have been reviewed in Yuan et al. (2023).

3. Analytical methods

In situ major element compositions of the olivine were determined using a JEOL JXA8100 electron probe microanalyzer (EPMA) at the Institute of Geology and Geophysics, Chinese Academy of Sciences (IGGCAS). The analytical conditions were accelerating voltage of 15 kV, 10 nA beam current, 5 μm beam spot and 10–30 s counting time on peak. Natural and synthetic minerals were used for standard calibration, and a program based on the ZAF procedure was used for matrix corrections. Analytical uncertainties for all elements analyzed were better than 1–2 % (RSD).

In situ trace elements analyses were performed on a 193 nm Coherent GeoLas Pro ArF Excimer laser coupled to an Element XR SF-ICP-MS instrument (LA-ICP-MS) at Metallogenetic Elements and Isotopes Laboratory at the IGGCAS. Ablation was performed using 44 μm diameter ablating spots at 5 Hz with an energy of 5 J/cm² for 60 s after measuring the gas blank for 30 s. A pre-ablation of three laser pulses was done prior to each analysis, and a 30 s washout between analyses was used. The carrier (He) and make-up gas (Ar) flows were optimized by a spot ablation of the NIST SRM 612 to obtain maximum signal intensities, while keeping ThO/Th at < 0.5 % and U/Th of 0.95–1.05. NIST SRM 610 reference glass was used as calibration material, and USGS BCR-2G as well as BIR-1G were analyzed for data quality (Li et al., 2023). Iron

(²⁸Si) was used as an internal standard. Data reduction was obtained using the GLITTER program. For most trace elements (> 0.10 ppm), the accuracy is better than ± 10 % with analytical precision (RSD) of ± 10 %.

4. Results

Major element compositions of olivine from different types of ores and sulfide-barren lherzolites are shown in Table S1. Olivine in the net-textured ores has forsterite (Fo) contents of 79.6 to 85.2 and Mn contents of 1270 to 2463 ppm (Fig. 2a). In the patchy net-textured ores, olivine has slightly low Fo values from 76.7 to 84.2, and high Mn contents between 1425 and 2865 ppm (Fig. 2b). In the disseminated ores, olivine has Fo values of 78.7 to 84.1 and Mn contents between 1440 and 2649 ppm. Notably, the olivine grains in the disseminated ores can be classified into two groups compositionally: one group with high Fo (>82) with relatively lower Mn contents, and the other with Fo lower than 82 (Fig. 2c). In the sulfide-barren lherzolites, olivine exhibits significantly lower Fo values from 69.7 to 81.8, with the highest Mn content ranging from 1495 to 3191 ppm (Fig. 2d). It is noteworthy that, while there is a general concordance between the elevation in both Fo content of olivine and interstitial sulfide concentrations, the specific ranges of Fo values associated with varying sulfide-silicate textures exhibit considerable overlap (Fig. 3a).

Trace element compositions determined by LA-ICP-MS and crystallization temperatures of olivine calculated using Al-in-olivine thermometer (De Hoog et al., 2010) are listed in Table S2. Olivine in the net-textured ores contains 49.3 to 209 ppm Al and 50.6 to 166 ppm Cr, with crystallization temperatures ranging from 803 to 1163 °C. In the patchy net-textured ores, olivine has 65.3 to 204 ppm Al and 49.2 to 184 ppm Cr, with crystallization temperatures from 871 to 1314 °C. Olivine in the disseminated ores features 50.8 to 225 ppm Al and 41.4 to 152 ppm Cr, with a broader temperature range of 821 to 1507 °C. In the sulfide-

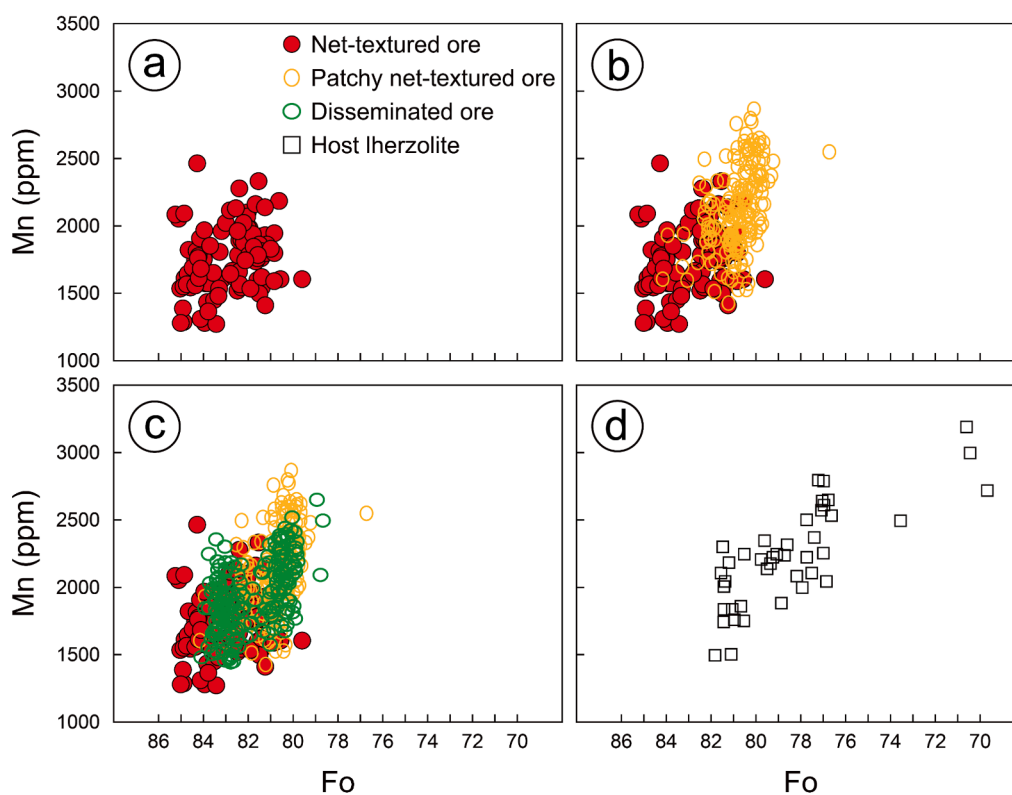


Fig. 2. Correlation diagrams of Fo vs. Mn content for olivine in (a–c) net-textured, patchy net-textured and disseminated ores and (d) sulfide-barren lherzolites from the orebody 1 of the Jinchuan deposit.

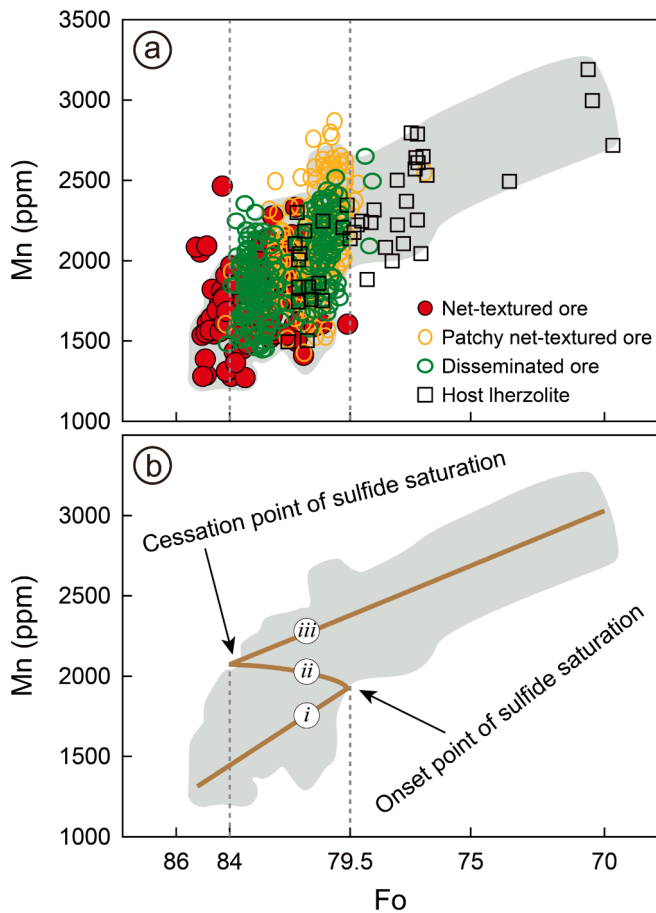


Fig. 3. (a) Fo vs. Mn correlation diagram for olivine from the orebody 1 of the Jinchuan deposit inferring (b) the putative magma evolution history of the Segment II, and the onset and the cessation points of sulfide saturation.

barren lherzolites, olivine contains 95.0 to 214 ppm Al and 68.0 to 194 ppm Cr, with the broadest temperature range from 929 to 1652 °C. These data present two primary clusters of olivine crystallization temperatures (Fig. 4a): a major group formed at lower temperatures, approximately between 800 and 1100 °C, and a minor group

crystallizing at significantly higher temperatures, ranging from about 1100 to 1600 °C.

5. Discussion

5.1. Relative timing of sulfide saturation

The deposition of sulfide in the Jinchuan intrusion has been hypothesized to be resulted from sulfide saturation either in a deep-seated magma chamber followed by subsequent ascent (e.g., Tang, 1993; De Waal et al., 2004; Ripley et al., 2005; Song et al., 2009), or as an in situ process within a magma conduit (Chai and Naldrett, 1992b). The Segment II, which harbors the orebody 1, formed from an initially crystal-free magma as inferred from the crystal size distributions of olivine (Mao et al., 2018). Consequently, olivine crystallized from the high-Mg basaltic parent magma (Chai and Naldrett, 1992b; Chen et al., 2009) would theoretically exhibit a decrease in Fo content along with a narrowing of the variability in contents of incompatible Mn ($D_{Mn}^{Ol/melt} = 0.63\text{--}0.77$; Jing et al., 2024) for a specific Fo value. Olivine grains from orebody 1 with Fo values exceeding 79.5 exhibit a pronounced breadth in Mn concentrations at any given Fo values (Fig. 3a). In contrast, when the Fo values drop below 79.5, there is a noticeable constriction in the variability of Mn contents (Fig. 3a). This discordance may suggest an influential role of sulfide saturation process on the magma composition. If the sulfide segregation has taken place in a deep chamber, then ensuing injection of the immiscible sulfide into the shallow chamber would likely not alter the Mn partitioning between olivine and the magma given the high incompatibility of Mn in sulfide melt ($D_{Mn}^{S/melt} = 0.10\text{--}0.40$; Li and Audétat, 2012). Therefore, the sulfide deposition in the orebody 1 was attributed to the in situ process within a magma conduit (Chai and Naldrett, 1992b).

5.2. Onset and cessation of in situ saturation

Due to the Fe-depletion and Mg- and Mn-enrichment caused by in situ sulfide saturation, co-crystallized olivine would exhibit an upward trend in both Fo value and Mn content. Upon the cessation of sulfide saturation, the silicate melt would resume normal differentiation, crystallizing olivine that exhibits a concomitant decrease in Fo value, and an increase in Mn content (Fig. 3b). Therefore, this in situ sulfide saturation is likely to produce a specific range of Fo values, each of which could correspond to a broad array of Mn contents (Fig. 3b), reflective of three

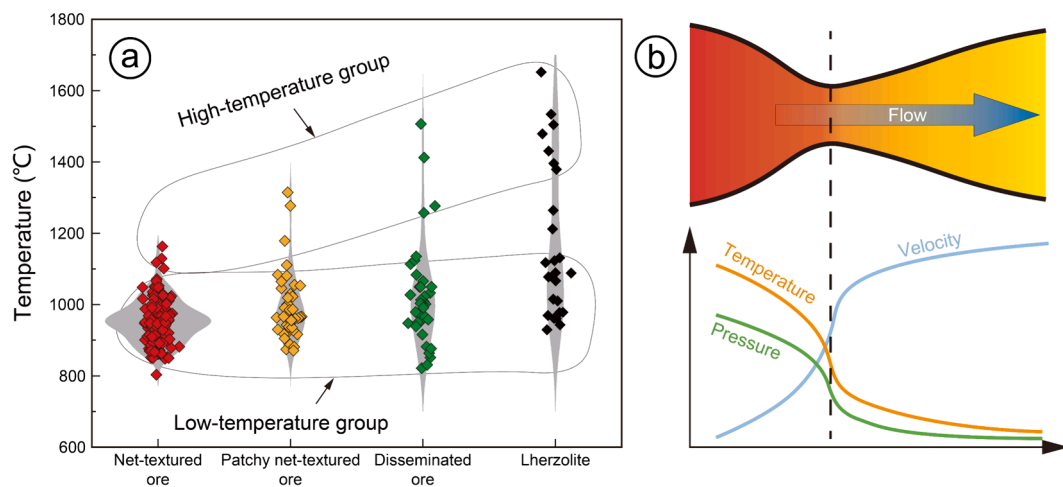


Fig. 4. (a) Crystallization temperatures of olivine from diverse sulfide-silicate textures are divided into two groups. The low-temperature group ranges from approximately 800 to 1100 °C, whereas the high-temperature group ranges from 1100 °C to 1650 °C. (b) Diagram of a de Laval nozzle showing an approximate increase in flow velocity with decreasing temperature and pressure along the flow, suggesting a similar effect on magma when it flows through a geometrically changing conduit resembling a de Laval nozzle.

distinct compositional stages: (i) prior to sulfide saturation, (ii) concurrent with sulfide saturation, and (iii) post-sulfide saturation.

Accordingly, the delineated olivine Fo-Mn dichotomy boundary at Fo79.5 (Fig. 3) therefore implies that sulfide saturation was initiated when the crystallizing olivine achieved a Fo value of 79.5. Consequently, the simultaneous gravitational olivine accumulation and sulfide percolation (Mao et al., 2018) would lead to the gradual occupation of interstitial spaces by siliceous magma towards the shallower levels, with a concomitant gradual reduction in sulfide content. This process fostered the progressive development of ores with net-textured and patchy net-textured to disseminated textures. Consequently, the maximal Fo value in olivine within the disseminated ore demarcates the termination of the sulfide saturation. Within the orebody 1 of the Jinchuan deposit, this termination is epitomized by olivine in disseminated ore with a peak Fo content of 84 (Fig. 3b), establishing a compositional range within which sulfide saturation initiated and ended at Fo values of 79.5 and 84, respectively.

5.3. Trigger of sulfide saturation

The deviations in sulfur isotopic compositions between the Jinchuan sulfide and the mix of mantle source and immediate country rocks indicate an en route sulfur incorporation (Xue et al., 2023). Furthermore, Xue et al. (2023) suggested that in situ carbonate contamination by the country rock (marble) has limited contribution to sulfide saturation. These findings rule out in situ crustal contamination as the primary contributor to in situ sulfide saturation and point to the potential involvement of other processes that may have played a role in the sulfide saturation within orebody 1. The crystallization temperatures of olivine indicate a decline in the thermal regime of the parent magma from approximately 1400–1500 °C to 800–1100 °C (Fig. 4a). Theoretically, this thermal reduction may be attributed to the contamination by the cold immediate carbonate country rock following magma emplacement, as evidenced by the O-Mg isotopic composition and CaO/SiO₂ ratio of the magma (Lehmann et al., 2007; Tang et al., 2018; Ding et al., 2021). However, the carbon isotopic compositions of the Jinchuan intrusion suggest that the effect of carbonate contamination was limited to the intrusion-country rock interface, which alone seems insufficient to account for the extensive sulfide saturation observed within the intrusion (Xue et al., 2023).

Geometrically, the Segment II, situated east to the fault F₁₆₋₁, presents a morphological transition from a narrow to a wide profile both horizontally (Fig. 1b) and vertically (Fig. 1c) on its western side. Coupled with the magma conduit identified on the western of Segment II (Dai et al., 2015), this configuration resembles a de Laval nozzle (Fig. 4b). Characteristically, such a nozzle comprises a contracting section leading to a minimum throat, followed by an expanding section. This unique structure caused a significant decrease in pressure and temperature (Fig. 4b, c) when the parent magma of the Jinchuan intrusion transited from west to east (Liu et al., 2020) through the throat.

A decrease in pressure is known to increase the value of SCSS (e.g., Mavrogenes and O'Neill, 1999; Li and Ripley, 2005, 2009; Ding et al., 2018). Therefore, reduction of pressure is not typically regarded as a precipitator for sulfide saturation. On the contrary, the decline in temperature has been both experimentally (e.g., Mavrogenes and O'Neill, 1999; Fortin et al., 2015; Ding et al., 2018) and empirically (Li and Ripley, 2005, 2009) shown to decrease the SCSS. Additionally, the temperature drop would significantly reduce SCSS, surpassing the SCSS increase that might be attributed to the decrease in pressure, as per the empirical equations proposed by Li and Ripley (2005, 2009). Furthermore, degassing of the hydrous parental magma of the Jinchuan intrusion (Yuan et al., 2023) due to depressurization could also contribute to a reduction in SCSS (e.g., Fortin et al., 2015). Collectively, these parameter changes, induced by a de Laval nozzle-shaped conduit, were likely to have precipitated the onset of sulfide saturation. This

hypothesis reflects that the geometric change of magma conduits may exert a significant influence on sulfide saturation, which is consistent with previous studies noting accumulations often occurring at points where conduits expansion or at the chamber entry (Naldrett, 1997; Li et al., 2000).

6. Conclusions

The varying correlations between Fo and Mn contents of olivine from different textures of ores and sulfide-barren lherzolites suggest an in situ sulfide saturation within orebody 1 of the Jinchuan deposit, which initiated with the crystallization of olivine with Fo value of 79.5 and ceased upon the crystallization of olivine with Fo value of 84. This saturation process coincides with a significant drop in olivine crystallization temperatures from ca. 1400–1500 °C to ca. 800–1100 °C. This temperature reduction likely occurred as the parent magma flowed through a geometrically changing conduit resembling a de Laval nozzle. This, along with resulting decreased oxygen fugacity and degassing of the hydrous parental magma, collectively led to the reduction of sulfur concentration at sulfide saturation and consequent in situ sulfide saturation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.oregeorev.2024.106121>.

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